

Sr. Data Analyst - Analytical Exercise

This exercise contains one main task (100 points) and one bonus task (20 points). The primary objective is to evaluate your proficiency in SQL query for data analysis and learn about how you conduct data exploration to derive actionable insights. Additionally, the bonus task is to understand how you apply real-world product experience to suggest analytical approaches to uncover insights for improving the user experience in your chosen product.

The bonus task is entirely voluntary and is an opportunity for you to showcase your problem-solving abilities. The total points for evaluation are capped at 100 and the points earned from the bonus task can supplement your overall score.

Feel free to explore more if you'd like, but this exercise should take about **60–75 minutes**. Please send your answer over to us within **3 business days**.

Task 1 Analysis (100 pts)

On a Monday morning, after your first sip of coffee, leadership has one question for you:

From Leadership

"We're not hitting our paid conversion targets and I want to understand what's driving it. We've been investing in paid acquisition but I'm not sure it's paying off. And I want to know if there's anything in how users behave early that has an impact on whether they convert."

The Data

You have access to "[thunkable.db](#)" ([Please download it here](#)), which is a SQLite database with a single table called events. Each row is one user action.

Table: events

Column	Description
user_id	Unique user identifier
event_time	Timestamp of the event
event_type	Type of action taken (see definitions below)
acquisition_source	How the user was acquired: organic, paid_search, referral, social
country	Country of the user

cohort_month	YYYY-MM of the user's signup month
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Event definitions

Event	What it means
sign_up	User created a Thunkable account
prompt_submitted	User typed their first prompt to generate an app
screen_previewed	User opened the live preview of their generated app
app_published	User published their app to a shareable link
collaborator_invited	User invited someone else to their project
upgrade_clicked	User clicked on a paid plan CTA
subscription_started	User converted to a paid plan

Use SQL to investigate the question. Feel free to speculate the business scenario and run with that. Please illustrate your speculations if needed.

Setup: Any SQL tool works: DB Browser for SQLite, DBeaver, Python sqlite3, anything.

Deliverables:

- SQL script as an appendix. Any read-only SQL syntax and function is accepted.
- Please provide a doc or slide deck of analysis report, 3-5 pages approximately, containing insights derived from the dataset to a Product Manager, readable for a non-technical audience. The report should include:
 - **Findings:** What is actually driving the conversion gap? Be specific about where in the funnel the problem sits.
 - **Recommendation:** What should the team do?
 - **Visualization:** 2–3 supporting charts, tables or any additional info that could help present key findings effectively.
 - **Flag (If any):** A section on anything in the data you would want to flag or verify before the team acts on your findings.
 - **Your thinking process:** One short paragraph on how you approached the problem. What did you look at first and why? What did you rule out? What would you dig into next with more time?

Optional Bonus Task: Onboarding Metric (20 pts)

Pick one product to analyze: Slack, Airtable or Thunkable. Use the resources below to familiarize yourself with the onboarding flow before answering. You don't need prior experience with any of them.

Resources:

- Slack (FAQPRIME: [Onboarding](#))
- Airtable (FAQPRIME: [Sign Up](#) and [Onboarding](#))
- Thunkable: If you opt for Thunkable, please kindly visit thunkable.com, sign up for a free account and try building something: enter a prompt describing an app, preview the result and attempt a 2nd prompt. (It should take you ~10 minutes)

Deliverables:

In one page, answer three questions:

- What is the one onboarding metric you'd prioritize and how would you measure it?
- Why does this metric matter and what outcome does it predict?
- If the metric looked unhealthy, what would you do next? What analysis would you run to investigate?

There's no right product to choose. We're looking for how you think about connecting a metric to a business outcome, not product knowledge.